

27/08/2025

Code-C



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Udyog Vihar, Gurugram, Haryana - 122015, Ph.+91-1244168300

MM : 720

Fortnightly Test for NEET-2026_RM(P3)_FT-02C

Time : 180 Min.

Topics Covered:**Physics:** Motion in a Plane, Laws of Motion**Chemistry:** Structure of Atom, Classification of Elements and Periodicity in Properties**Botany:** Cell Cycle & Cell Division, The Living World, Biological Classification-I: (Upto thermoacidophiles)**Zoology:** Biomolecules-II: (Proteins, types & functions, Lipids, Nucleic acids, Enzymes, Cofactors), Breathing & Exchange of Gases-I: (Upto mechanism of breathing)**General Instructions :**

Duration of Test is 3 hrs.

The Test consists of 180 questions. The maximum marks are 720.

There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.

Each question carries +4 marks. For every wrong response, -1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.

Use blue/black ballpoint pen only to darken the appropriate circle.

Mark should be dark and completely fill the circle.

Dark only one circle for each entry.

Dark the circle in the space provided only.

Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

PHYSICS

1. Two cars A and B are moving in uniform circular motion in concentric circles of radii r_A and r_B with speeds v_A and v_B respectively. Their time period of rotation is same. The ratio of angular speed of car A to that of car B will be

- (1) 1 : 1
(2) $r_A : r_B$
(3) $v_A : v_B$
(4) $r_A v_B : r_B v_A$

2. If $\vec{A} = 2\hat{i} + 3\hat{j}$ and $\vec{B} = 3\hat{i} + 4\hat{j}$, then the vector having magnitude same as magnitude of $\vec{B}$ and parallel to $\vec{A}$ is

- (1) $\frac{5}{\sqrt{13}}(2\hat{i} + 3\hat{j})$
(2) $\frac{2\hat{i} + 3\hat{j}}{\sqrt{13}}$
(3) $\frac{5}{\sqrt{7}}(3\hat{i} + 2\hat{j})$
(4) $\frac{\hat{i} + 2\hat{j}}{\sqrt{17}}$

3. A swimmer wants to cross the river in shortest possible time. The angle θ made by the swimmer with flow of river is

- (1) $\theta = 0^\circ$
(2) $\theta > \frac{\pi}{2}$
(3) $\theta = \frac{\pi}{2}$
(4) $0 < \theta < \frac{\pi}{2}$

4. Rain is falling in vertically downward direction with speed $10\sqrt{3}$ km/h. If wind is blowing in eastward direction with speed 10 km/h, then the velocity of rain observed by a person standing on the ground is

- (1) 20 km/h
(2) 10 km/h
(3) $10\sqrt{3}$ km/h
(4) $10(\sqrt{3} - 1)$ km/h

5. A particle is projected with velocity 40 m/s at an angle 60° with horizontal at $t = 0$. The time when the velocity of particle becomes parallel to the horizontal is ($g = 10 \text{ m/s}^2$)

- (1) $2\sqrt{3} \text{ s}$
(2) $\frac{4}{\sqrt{3}} \text{ s}$
(3) $4\sqrt{3} \text{ s}$
(4) $\frac{4}{3} \text{ s}$

5 bullets/sec
mass = 20g
100 m/s

6. A projectile is thrown at an angle θ with the horizontal with speed u . After some time, it makes an angle α with the horizontal. The speed of the projectile at that instant is

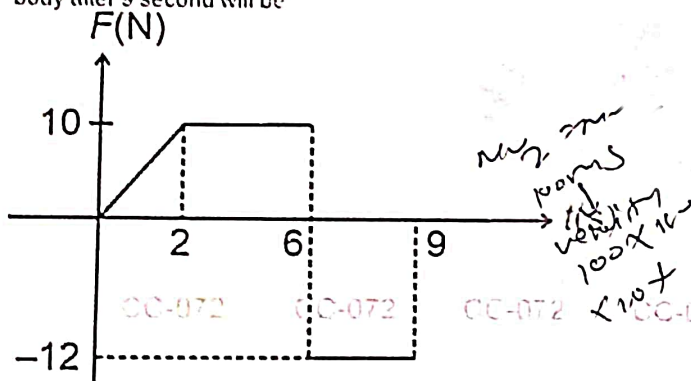
- (1) $u(\cos \theta + \cos \alpha)$
(2) $u(\cos \theta - \cos \alpha)$
(3) $\frac{u \cos \theta}{\cos \alpha}$
(4) $\frac{u \sin \theta}{\cos \alpha}$

$u \cos \theta = u \cos \alpha$
 $\frac{u \cos \theta}{\cos \alpha} \times \frac{1}{\cos \alpha}$

7. If $\vec{A} = 4\hat{i} + 2\hat{j} + 5\hat{k}$ and $\vec{B} = -2\hat{j} - 5\hat{k}$, the angle made by $\vec{A} + \vec{B}$ with positive y-axis is

- (1) 45°
(2) 60°
(3) Zero
(4) 90°

8. A body of mass 1 kg is moving on a straight line with speed 3 m/s. A force starts acting on it along its line of motion which is shown by force-time graph below. The speed of the body after 9 second will be

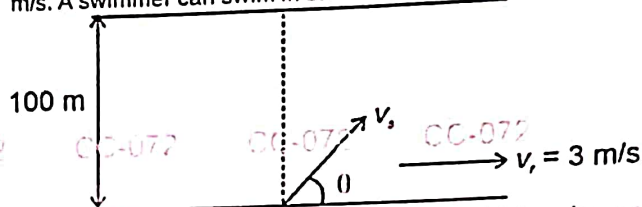


- (1) 17 m s^{-1}
(2) 14 m s^{-1}
(3) 21 m s^{-1}
(4) 13 m s^{-1}

9. A machine gun fires 5 bullets per second and the mass of each bullet is 20 g. If 100 m/s is the speed of each bullet, then the force exerted on the machine gun is

- (1) 10 N
(2) 20 N
(3) 8 N
(4) 15 N

10. A river has a width of 100 m and it is flowing with speed 3 m/s. A swimmer can swim in still water with 5 m/s.



As shown in the figure, θ is the angle at which swimmer swims with respect to the river flow. Match the columns and tick the correct option.

Column I

Column II

- | | |
|---|-----------|
| a. Value of θ (in degrees) for zero drift | (i) 60 |
| b. Value of θ (in degrees) for minimum time to cross the river | (ii) 25 |
| c. Value of drift (in m) in case of minimum time | (iii) 127 |
| d. Value of time (in s) in case of zero drift | (iv) 90 |
| | (v) 45 |

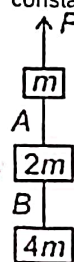
(1) a(iii), b(v), c(i), d(ii)

(2) a(iv), b(v), c(i), d(ii)

(3) a(iii), b(iv), c(i), d(ii)

(4) a(iii), b(iv), c(ii), d(i)

11. Three blocks with masses m , $2m$ and $4m$ are connected by strings A and B as shown in figure. After an upward force F is applied on block m , the masses are moving upward at constant speed v . The tension in the string B will be



- (1) $7mg$
(2) $6mg$
(3) Zero
(4) $4mg$

12. A block of mass 2 kg lies on a horizontal surface in a truck. The coefficient of friction between the block and surface is 0.5. If the acceleration of the truck is 4 m/s^2 , then friction force acting on the block will be ($g = 10 \text{ m/s}^2$)

(1) 10 N
(2) 8 N
(3) 6 N
(4) Zero

13. A ship A is moving westwards with a speed of 10 km/h and a ship B at 100 km south of A, is moving northwards with a speed of 10 km/h. The time after which the distance between them becomes shortest is

(1) $10\sqrt{2} \text{ h}$
(2) $\sqrt{2} \text{ h}$
(3) 5 h
(4) $5\sqrt{2} \text{ h}$

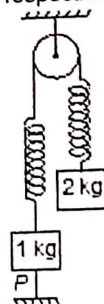
14. A particle moves in x-y plane according to the law $\vec{v} = 3y^2\hat{i} + 2x\hat{j}$. The equation of trajectory is

(1) $\frac{x^3}{3} + \frac{y^4}{4} = \text{constant}$
(2) $\frac{y^4}{3} - \frac{x^3}{4} = \text{constant}$
(3) $\frac{y^4}{4} - \frac{x^3}{3} = \text{constant}$
(4) $x^2 - y^3 = \text{constant}$

15. A fighter plane is flying horizontally at an altitude of 4500 m with speed 720 km/h. At what angle of sight w.r.t. horizontal when the target is seen, should the pilot drop the bomb in order to strike the target? ($g = 10 \text{ m/s}^2$)

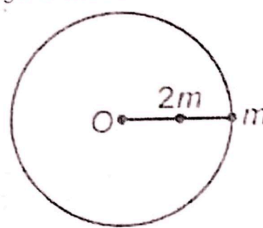
(1) 30°
(2) 45°
(3) 60°
(4) 37°

16. In given figure pulley and strings are massless. If string P is cut, the acceleration of blocks 1 kg and 2 kg after cutting are respectively [Initially system is in equilibrium]



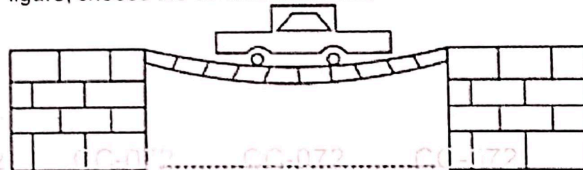
(1) g upward, zero
(2) Zero, g upward
(3) g downward, zero
(4) Zero, g downward

17. Two masses $2m$ and m are whirled in a horizontal circle. The radius of path of $2m$ is r and of m is $2r$ as shown in figure. Then the ratio of tension of two strings are



(1) 2 : 1
(2) 1 : 1
(3) 3 : 4
(4) 4 : 3

18. For a motor car moving over a concave bridge as shown in figure, choose the correct statement.



(1) Motor car moving over bridge is lighter than the same car resting on the same bridge.
(2) Motor car moving over bridge is heavier than the same car resting on the same bridge.
(3) Weight will be same in both the cases.
(4) Data insufficient

180
18
162
162
648

19. A particle is moving on a circle of radius 2 m. Its speed (v) varies with time t as $v = \alpha t^2$, where α is a constant. The ratio of radial acceleration to tangential acceleration of the particle at $t = 2$ s is

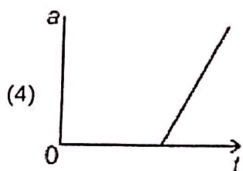
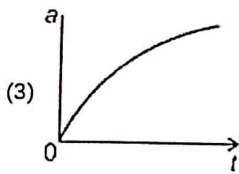
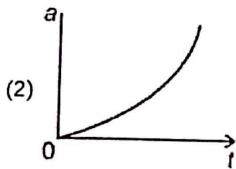
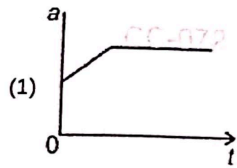
(1) α
(2) 8
(3) 2α
(4) 4

$$V = \alpha t^2 \quad r = 2\text{m} \quad S = \frac{1}{2} \alpha t^2$$

$$t = 2\text{s}$$

$$V = u + at$$

20. A block is placed on a rough horizontal surface. A time dependent horizontal force $F = 2t$ acts on the block. The acceleration - time ($a - t$) graph of block is best represented by



21. If resultant of two unit vectors $\hat{a}$ and $\hat{b}$ is $\sqrt{3}$ in magnitude, then the magnitude of difference of these unit vectors is

(1) Zero
(2) 1
(3) $\sqrt{3}$
(4) $\sqrt{2}$

22. Consider the two statements related to circular motion in usual notations.

Statement I : In non-uniform circular motion $\vec{\omega}$, $\vec{v}$ and $\vec{a}$ are always mutually perpendicular.

Statement II : In uniform circular motion $\vec{v}$, $\vec{r}$ and $\vec{a}$ are always mutually perpendicular.

(1) Both statements are true
(2) Both statements are false
(3) Statement I is true but statement II is false
(4) Statement II is true but statement I is false

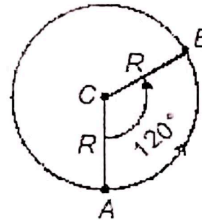
23. A particle has an initial velocity of $3\hat{i} + 2\hat{j}$ and acceleration of $0.1\hat{i} + 0.1\hat{j}$. Its speed after 10 s is

(1) 10 units
(2) 5 units
(3) 7 units
(4) 8.5 units

24. The position of a particle is expressed as $\vec{r} = (2t^2\hat{i} + 6t\hat{j})$ m, where t is time in second. The direction of velocity with x-axis at time $t = 2$ second is

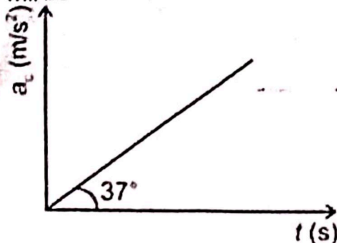
(1) 45°
(2) 30°
(3) 53°
(4) 37°

25. A particle is moving on a circular path of radius $R = \frac{10}{\pi}$ m with constant speed 10 m/s. If it moves from point A to point B as shown in figure, then magnitude of average acceleration of the particle during its journey will be



(1) $10\sqrt{3} \text{ m/s}^2$
(2) 10 m/s^2
(3) $5\sqrt{3} \text{ m/s}^2$
(4) 5 m/s^2

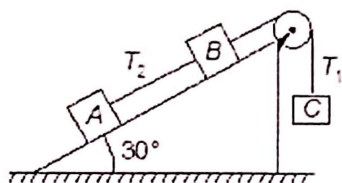
26. Centripetal acceleration (a_c) of a particle moving in a circle of radius 12 m varies with time t as shown in the figure. If particle starts from rest, then speed of the particle after 1 s will be



- (1) 2 m/s
(2) 3 m/s
(3) 4 m/s
(4) 5 m/s
27. A body is projected with same speed at two different angles, covers the same horizontal distance. If T_1 and T_2 are respective time of flights, then horizontal distance is equal to

- (1) gT_1T_2
(2) $2gT_1T_2$
(3) $\frac{gT_1T_2^2}{T_1+T_2}$
(4) $\frac{g}{2}T_1T_2$

28. Three blocks A, B and C of masses 5 kg, 5 kg and 10 kg respectively are connected with two light strings one of them is passing over an ideal pulley. The system is released from rest on smooth inclined plane of inclination 30° as shown in figure. The value of tensions $\frac{T_1}{T_2}$ is



- (1) 2
(2) $\frac{\sqrt{3}}{2}$
(3) $\frac{1}{2}$
(4) $\frac{2}{\sqrt{3}}$

29. A 50 kg man stuck in flood is being lifted vertically by an army helicopter with the help of light rope which can bear a maximum tension of 70 kg-wt. The maximum acceleration with which helicopter can rise so that rope does not break, is ($g = 9.8 \text{ ms}^{-2}$)

$$m = 50 \text{ kg} \quad 70 \text{ kg-wt}$$

- (1) 4.00 ms^{-2}
(2) 3.92 ms^{-2}
(3) 3.52 ms^{-2}
(4) 3.00 ms^{-2}

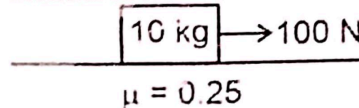
30. A batsman hits back a ball of mass 0.2 kg straight in the direction of bowler without changing its initial speed of 10 m s^{-1} . If the ball moves linearly, then impulse imparted to ball is

- (1) 2 N s
(2) 5 N s
(3) 8 N s
(4) 4 N s

31. A force acts for 26 s on a body of mass 13 kg, starting from rest, after which the force ceases and then body describes 75 m in the next 15 s. The value of force will be

- (1) 1.5 N
(2) 2 N
(3) 2.5 N
(4) 3 N

32. A block of mass 10 kg is under the action of horizontal force 100 N, placed on a rough horizontal surface ($\mu = 0.25$). The acceleration of block is

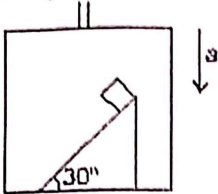


- (1) 10 m/s^2
(2) 2.5 m/s^2
(3) 7.5 m/s^2
(4) 7.5 m/s^2

33. A gramophone record is revolving with an angular velocity ω . A coin is placed at a distance $2r$ from the centre of the record. The coefficient of static friction is μ . The coin will revolve with the record if

- (1) $\omega \leq \sqrt{\frac{3\mu g}{r}}$
(2) $\omega \leq \sqrt{\frac{2\mu g}{r}}$
(3) $\omega \leq \sqrt{\frac{\mu g}{r}}$
(4) $\omega \leq \sqrt{\frac{\mu g}{2r}}$

34. A block of mass 2 kg is released from the top of a smooth fixed wedge inside the elevator which is moving downward with an acceleration $a = 5 \text{ m/s}^2$ as shown in the figure. The value of acceleration of block w.r.t wedge will be ($g = 10 \text{ m/s}^2$)



- (1) 10 m/s^2
 (2) 5 m/s^2
 (3) 2.5 m/s^2
 (4) $5\sqrt{2} \text{ m/s}^2$
35. A thin circular ring of radius R , with a bead free to slide, rotates about its vertical diameter with an angular frequency ω . The angle made by the radius vector joining centre to bead with the vertically upward direction, when ring rotates with angular frequency $\omega = \sqrt{\frac{g}{R}}$ is (Neglect friction and consider that bead is initially at lowest position).

- (1) 180°
 (2) 135°
 (3) 90°
 (4) 45°
- Handwritten notes for 35: $\omega R = 5 \text{ m/s}$, Radius R , $g = 10 \text{ m/s}^2$, $\sin \theta = \frac{\omega R}{g} = \frac{5}{10} = \frac{1}{2}$, $\theta = 30^\circ$, $\theta = 180 - 30 = 150^\circ$*

36. A ball is projected under gravity with velocity 20 m/s at an angle θ with the horizontal such that horizontal range is equal to the maximum height of projectile. The range of projectile is ($g = 10 \text{ ms}^{-2}$)

- (1) 160 m
 (2) 320 m
 (3) $\frac{160}{17} \text{ m}$
 (4) $\frac{320}{17} \text{ m}$
- Handwritten notes for 36: $uH = \tan \theta R$, $V = 20 \text{ m/s}$, Horizontal Range $= \theta$, $uH = R$, $u = \tan \theta$, $u = \tan \theta$, $\frac{u}{1} \sin \theta = \frac{1}{2}$ height*

37. Assertion (A): For a given mass, greater the speed, the greater is the opposing force needed to stop the body in a certain time.

Reason (R): The greater the change in the momentum in a given time, the greater is the force that needs to be applied.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
 (3) (A) is false but (R) is true
 (4) (A) is true but (R) is false

38. A particle is at rest at origin. Two perpendicular accelerations $a_x = 1 \text{ m/s}^2$ and $a_y = 3 \text{ m/s}^2$ start acting on it, then the trajectory of motion of the particle will be

- (1) Circle
 (2) Ellipse
 (3) Straight line
 (4) Parabola

39. The angular speed of a motor wheel is increased from 20 rps to 52 rps in 16 second. How many revolutions does engine make during this time? (Assuming uniform angular acceleration).

- (1) 432
 (2) 618
 (3) 912
 (4) 576

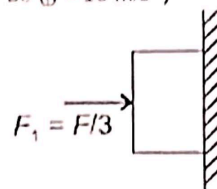
Handwritten note for 39: 20 rps to 52 rps

40. Two forces 6 N and 4 N are at an angle θ such that the resultant is R . The first force is now increased to 12 N while second remaining same, then the resultant becomes $2R$. The value of θ is

- (1) $\frac{\pi}{3}$
 (2) $\frac{\pi}{2}$
 (3) $\frac{2\pi}{3}$
 (4) $\frac{3\pi}{4}$

Handwritten notes for 40: 6N and 4N, Resultant R, 12N and 4N, Resultant 2R, $\theta = 120^\circ$

41. The minimum force required to keep a block of mass 1 kg at rest against rough vertical wall having coefficient of friction between wall and block ($\mu = 0.5$) is F . If a force $\frac{F}{3}$ is applied on block as shown in figure, then acceleration of block will be ($g = 10 \text{ m/s}^2$)



- (1) $\frac{10}{3} \text{ m/s}^2$
 (2) 10 m/s^2
 (3) 20 m/s^2
 (4) $\frac{20}{3} \text{ m/s}^2$

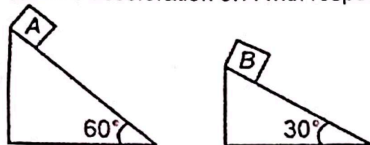
Handwritten notes for 41: $F = \frac{1}{3}$, mass = 1 kg, $F = \mu N$, $\mu = 0.5$, $F/3$

Handwritten note for 41: 5 m/s^2 , $g = 10 \text{ m/s}^2$

Handwritten note for 41: 2 kg, $a = 5 \text{ m/s}^2$, The Value of Acceleration

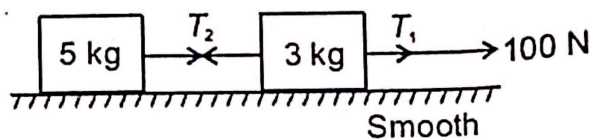
Handwritten notes at bottom: $\sin 30 = 1/2$, $\cos 30 = \sqrt{3}/2$, $\frac{1}{2} = \frac{1}{2} \times \frac{1}{2}$, $\frac{1}{2} = \frac{1}{2} \times \frac{1}{2}$, $\cos \theta = \frac{1}{2}$

42. Two fixed frictionless inclined plane making an angle 30° and 60° with the horizontal are shown in the figure. Two blocks A and B are placed on the two planes. What is the relative acceleration of A with respect to B?



- (1) 4.9 ms^{-2} in horizontal direction
 (2) 9.8 ms^{-2} in vertical direction
 (3) Zero
 (4) 4.9 ms^{-2} in vertical direction
43. There are two inclined surfaces of equal length (L) and same angle of inclination 45° with the horizontal. One of them is rough and the other is perfectly smooth. A given body takes 2 times as much time to slide down on rough surface than on the smooth surface. The coefficient of kinetic friction (μ_k) between the object and the rough surface is close to
- (1) 0.25
 (2) 0.40
 (3) 0.5
 (4) 0.75

44. As shown in figure, find the tension T_2 .



- (1) 62.5 N
 (2) 22.5 N
 (3) 77.5 N
 (4) Zero
45. The path of projectile is represented by $y = ax - bx^2$, then match the columns.

	Column I		Column II
i.	Range	(a)	a
ii.	Maximum height	(b)	$\frac{a}{b}$
iii.	Tangent of angle of projection	(c)	$\frac{a^2}{4b}$

- (1) i(c), ii(b), iii(a)
 (2) i(b), ii(c), iii(a)
 (3) i(a), ii(c), iii(b)
 (4) i(a), ii(b), iii(c)

$$y = ax - bx^2$$

Range

CHEMISTRY

46. Consider the following two statements

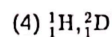
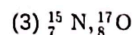
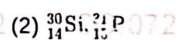
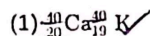
Statement I : Bohr's theory was able to explain the splitting of spectral lines in the presence of magnetic field or an electric field.

Statement II : Bohr's theory could not explain the ability of atoms to form molecules by chemical bonds.

In the light of above statements, choose the correct option.

- (1) Both statement I and statement II are correct
 (2) Statement I is correct and statement II is incorrect
 (3) Statement I is incorrect and statement II is correct
 (4) Both statement I and statement II are incorrect

47. Which of the following pairs can be called as isobars?



48. If a 20 watt bulb emits monochromatic light of wavelength 1980 nm then the number of photons emitted per second by the bulb will be ($h = 6.6 \times 10^{-34} \text{ Js}$)

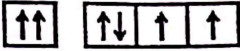
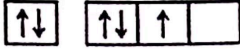
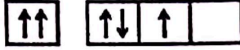
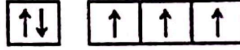
- (1) 6×10^{21}
 (2) 4×10^{18}
 (3) 2×10^{20}
 (4) 8×10^{22}

49. Given below are two statements one is assertion (A) other is reason (R)
Assertion (A): First ionization energy for nitrogen is higher than that of oxygen.
Reason (R): On moving left to right across a period effective nuclear charge decreases.
 In the light of above statements choose the correct answer.
 (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
 (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
 (3) Assertion is true statement but Reason is false
 (4) Both Assertion and Reason are false statements
50. Pair of ions having equal number of unpaired electrons is
 (1) Cr^{2+} and Fe^{3+}
 (2) Ni^{2+} and Fe^{2+}
 (3) Cr^{2+} and Fe^{2+}
 (4) Mn^{4+} and Ni^{2+}
51. Maximum number of electrons in a subshell with $n = 2$ and $l = 1$ is
 (1) 2
 (2) 6
 (3) 10
 (4) 14
52. Consider the following statements
 a. Azimuthal quantum number defines the three dimensional shape of orbital
 b. Principal quantum number determines the size of an orbital
 c. Third shell of He^+ has 9 degenerate orbitals
 The correct statements are
 (1) a and b only
 (2) b and c only
 (3) a and c only
 (4) a, b and c
53. For multielectronic species, the correct order of energy of the given orbitals is
 (1) $3d > 4p > 5s > 5p$
 (2) $4p > 3d > 5p > 5s$
 (3) $5p > 5s > 4p > 3d$
 (4) $5s > 4p > 3d > 5p$
54. Among group-16 and group-17 elements, the elements having minimum and maximum negative electron gain enthalpy respectively are
 (1) O and Cl
 (2) Po and F
 (3) O and F
 (4) S and Cl
55. Halogens are present in which group of periodic table?
 (1) 14
 (2) 15
 (3) 16
 (4) 17
56. If a particular station of All India Radio, New Delhi broadcasts on a frequency of 1500 kHz (kilo hertz) then the wavelength of the electromagnetic radiation emitted by the transmitter will be
 (1) 400 m
 (2) 200 m
 (3) 100 m
 (4) 300 m
57. Consider the following phenomena.
 (a) Diffraction
 (b) Interference
 (c) Black body radiation
 (d) Photoelectric effect
 The phenomenon that cannot be explained by wave nature of electromagnetic radiation are
 (1) a and b only
 (2) c and d only
 (3) a and c only
 (4) b and d only
58. On the basis of Rutherford model of atom radius of nucleus is about
 (1) 10^{-10} m
 (2) 10^{-12} m
 (3) 10^{-15} m
 (4) 10^{-20} m
59. Orbital angular momentum of electron present in 3d orbital is
 (1) $\sqrt{2} \hbar$
 (2) $2\sqrt{3} \hbar$
 (3) $\sqrt{6} \hbar$
 (4) $\sqrt{3} \hbar$

$$n - (n - 1)$$

$$6 \times 5$$

60. Maximum number of visible lines when an electron returns from 6th orbit to ground state in H spectrum will be
- 5
 - 4
 - 6
 - 3
61. The number of protons, neutrons and electrons in $^{175}_{71}\text{Lu}$, respectively, are
- 71, 71 and 104
 - 175, 104 and 71
 - 71, 104 and 71
 - 104, 71 and 71
62. Which of the following statements are true regarding modern periodic table?
- s-block elements of the same group have similar valence shell electronic configuration.
 - Metals comprise higher percentage than non-metals of the known elements.
 - The physical and chemical properties of elements vary periodically with their atomic numbers.
- Choose the correct answer from the options given below.
- (a) and (b) only
 - (a) and (c) only
 - (b) and (c) only
 - (a), (b) and (c)
63. Which among the following processes is an endothermic process?
- $\text{F(g)} \rightarrow \text{F}^-(\text{g})$
 - $\text{Br(g)} \rightarrow \text{Br}^-(\text{g})$
 - $\text{O}^-(\text{g}) \rightarrow \text{O}^{2-}(\text{g})$
 - $\text{O(g)} \rightarrow \text{O}^-(\text{g})$
64. Consider the following statements about cathode rays:
- They are negatively charged particles
 - The characteristics of cathode rays do not depend upon the material of electrodes used.
 - They start from anode and move towards cathode.
- The correct statements are:
- (a) and (b) only
 - (b) and (c) only
 - (a) and (c) only
 - (a), (b) and (c)
65. Given below are two statements: One is labelled as Assertion (A) and other is labelled as Reason (R).
- Assertion(A) : Na^+ and Al^{3+} are iso-electronic but ionic radius of Na^+ is less than that of Al^{3+} .
- Reason(R) : The magnitude of effective nuclear charge on the outer shell electrons in Na^+ is greater than that in Al^{3+} .
- In the light of above statements, choose the most appropriate answer from the options given below.
- Both Assertion & Reason are true and the reason is the correct explanation of the assertion
 - Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
 - Assertion is true statement but Reason is false
 - Both Assertion and Reason are false statements
66. Orbital in which electron density is symmetrically distributed is
- 1s
 - 2p_x
 - 3p_y
 - 4d_{xy}
67. The energies E_1 and E_2 of two radiations are 25 eV and 100 eV respectively. The relation between their wavelength i.e. λ_1 and λ_2 will be
- $\lambda_1 = 4\lambda_2$
 - $\lambda_2 = 4\lambda_1$
 - $\lambda_1 = 2\lambda_2$
 - $\lambda_2 = 2\lambda_1$
68. The energy of an electron in a Bohr's orbit of hydrogen atom is - 1.51 eV. The radius of orbit is
- 4.76 Å
 - 2.12 Å
 - 1.06 Å
 - 1.59 Å
69. The electronic configuration of chromium atom (atomic number = 24) is
- $[\text{Ar}]4s^23d^4$
 - $[\text{Ar}]5s^14d^5$
 - $[\text{Ar}]4s^13d^5$
 - $[\text{Ar}]4s^14d^5$

70. Eka – aluminium and Eka - silicon in Mendeleev's periodic table respectively are
- Ga, Ta
 - Ga, Sn
 - Ge, Ga
 - Ga, Ge
71. Maximum kinetic energy of photoelectrons emitted when a light of frequency 1.1×10^{12} Hz is irradiated on a metal surface whose threshold frequency is equal to 1.0×10^{11} Hz, is ($h = 6.6 \times 10^{-34}$ J s)
- 6.6×10^{-25} J
 - 1.8×10^{-21} J
 - 5.1×10^{-20} J
 - 6.6×10^{-22} J
72. If radius of second orbit of Li^{2+} is x Å then radius of the first orbit of He^+ will be
- $3x$ Å
 - $\frac{2}{3}x$ Å
 - $\frac{5}{8}x$ Å
 - $\frac{3}{8}x$ Å
73. The de-Broglie wavelength of a moving object having mass 5 g and velocity 200 m/s will be ($h = 6.63 \times 10^{-34}$ Js)
- 6.63×10^{-37} m
 - 6.63×10^{-32} m
 - 6.63×10^{-34} m
 - 6.63×10^{-36} m
74. Number of radial and angular nodes present in $4d$ orbitals respectively are
- 2 and 2
 - 2 and 1
 - 1 and 2
 - 3 and 2
75. According to the Bohr Theory, which of the following transitions in the hydrogen atom will give rise to the least energetic photon?
- $n = 6$ to $n = 5$
 - $n = 5$ to $n = 3$
 - $n = 6$ to $n = 1$
 - $n = 5$ to $n = 4$
76. Which of the following orbital diagrams violates both Pauli's exclusion principle as well as Hund's rule?
- 
 - 
 - 
 - 
77. Match the atomic numbers given in column I with their IUPAC official names given in column II.
- | Column I | Column II |
|----------|-------------------|
| a. 101 | (i) Rutherfordium |
| b. 103 | (ii) Mendelevium |
| c. 104 | (iii) Lawrencium |
| d. 108 | (iv) Hassium |
- Choose the correct match from the options given below.
- a(iii), b(ii), c(iv), d(i)
 - a(i), b(ii), c(iii), d(iv)
 - a(ii), b(iii), c(i), d(iv)
 - a(iii), b(ii), c(i), d(iv)
78. Identify the group and period respectively, for an element with atomic number 113.
- 13 and 6
 - 13 and 7
 - 12 and 7
 - 14 and 6
79. Which among the following is not a representative element?
- Mg
 - Ga
 - P
 - Zn
80. Maximum number of electrons in an atom having $n + l + m = 3$ is
- 6
 - 2
 - 8
 - 4
81. Which of the following is not permissible arrangement of electrons in an atom?
- $n = 4, l = 1, m = 0, s = +\frac{1}{2}$
 - $n = 5, l = 2, m = 2, s = -\frac{1}{2}$
 - $n = 5, l = 5, m = 1, s = -\frac{1}{2}$
 - $n = 4, l = 3, m = 2, s = +\frac{1}{2}$

	Column I	Column II
(a)	Copper	(i) Non-metal
(b)	Fluorine	(ii) Transition Metal
(c)	Silicon	(iii) Lanthanoid
(d)	Cerium	(iv) Metalloid

Identify the correct match

- (1) (a)-(i) (b)-(ii) (c)-(iii) (d)-(iv)
(2) (a)-(ii) (b)-(iv) (c)-(i) (d)-(iii)
~~(3) (a)-(ii) (b)-(i) (c)-(iv) (d)-(iii)~~
(4) (a)-(iv) (b)-(iii) (c)-(i) (d)-(ii)

91. Identify the correct sequence of events occur during meiosis.
- Dissolution of synaptonemal complex.
 - Exchange of genetic material between two homologous chromosome.
 - Formation of complex by a pair of synapsed homologous chromosomes.
 - Splitting of centromere of each chromosome
 - Separation of homologous chromosomes.

(1) C - D - A - E - B

(2) C - A - B - E - D

(3) C - B - A - D - E

(4) ~~C - B - A - E - D~~

CBAED

92. Read the following statements and select the correct option.

Statement A : Liquid endosperm of coconut is an example of syncytium.

Statement B : Meiosis results in formation of haploid daughter cells which are identical to each other.

(1) Only A is incorrect

(2) Only B is incorrect

(3) Both A and B are correct

(4) Both A and B are incorrect

93. How many bivalents can be observed if there are total 36 chromosomes in a diploid cell?

(1) 18

(2) 9

(3) 36

(4) 2

94. Match the given columns and choose the correct option.

	Column I		Column II
2	a. Zygotene	(i)	Formation of recombination nodule
4	b. Diakinesis	(ii)	Pairing of homologous chromosomes
1	c. Pachytene	(iii)	Desynapsis
3	d. Diplotene	(iv)	Terminalisation of chiasmata

(1) a(iii), b(i), c(iv), d(ii)

(2) a(ii), b(iii), c(i), d(iv)

(3) a(iii), b(ii), c(iv), d(i)

(4) a(ii), b(iv), c(i), d(iii)

95. Meiosis II is similar to mitosis, but not an exact copy of mitosis because

(1) It always occurs in diploid cells

(2) There is no DNA replication just prior to meiosis II

(3) Daughter cells, formed after meiosis II are similar to parent cell

(4) It is reductional division phase of meiosis

96. How many mitotic divisions are required to produce 64 cells from one cell?

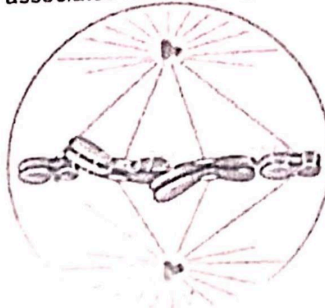
(1) 64

(2) 16

(3) 63

(4) 8

97. Select the option which correctly mentions an event associated with the stage of cell cycle shown in the figure.



(1) Chromosome losing their identity as discrete elements

(2) Movement of chromosome towards the opposite poles

(3) Initiation of the assembly of mitotic spindles

(4) Chromosome aligned at the equator with the help of spindle fibres

98. Read the following statements and choose the correct option.

Assertion : During zygotene stage, chromosomes start pairing together and this process is called terminalisation.
Reason : Anaphase-I begins with simultaneous splitting of centromeres of each chromosome.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

99. Dyad of cells produced in reductional division

(1) Are formed just after diakinesis

(2) Have haploid nuclei

(3) Have bivalent chromosomes

(4) Undergo crossing over

100. How many of the given events are correct w.r.t S-phase of cell cycle
- Duplication of centriole
 - Tubulin protein synthesis
 - Duplication of most of the organelle
 - Synthesis of nucleotides
 - Synthesis of histones
- 1
 - 2
 - 3
 - 4
101. State true (T) or false (F) for the given statements and choose the correct option.
- Meiosis increases the genetic variability
 - Meiosis maintains the chromosome number from generation to generation
 - Mitosis maintains the size of the cell
- A-(T), B-(T), C-(F)
 - A-(T), B-(T), C-(T)
 - A-(F), B-(F), C-(T)
 - A-(T), B-(F), C-(T)
102. In animal cell, cytokinesis is achieved by
- Furrow formation with the help of microfilaments
 - Cell plate formation with the help of microtubules
 - Appearance of furrow with the help of microtubules
 - Cell plate formation with the help of myosin protein
103. Select the correct option w.r.t. event that leads to formation of syncytium.
- Karyokinesis not followed by disappearance of nuclear membrane
 - Karyokinesis not followed by cytokinesis
 - Cytoplasm division without DNA replication
 - Cytokinesis followed by karyokinesis
104. How many asters together with spindle fibres forms mitotic apparatus?
- 1
 - 2
 - 3
 - 4
105. Double metaphasic plate is formed during
- Metaphase I
 - Metaphase II
 - Metaphase of mitosis
 - Anaphase I
106. Select the correct match.
- Prophase – Gradual uncoiling of chromosomes
 - Anaphase – Formation of interzonal fibres
 - Telophase – Congression of chromosome occurs
 - Metaphase – Nuclear membrane reassemble
107. Average duration of cell cycle of a typical human cell in culture is
- 90 minutes
 - 24 hours
 - 90 hours
 - 24 minutes
108. Some cells of adult animals like nerve cells
- Are metabolically inactive
 - Show continuous proliferation
 - Enter the quiescent stage
 - Replicate its DNA during M-phase
109. Interval between mitosis and initiation of DNA replication is also called
- Post-mitotic gap phase
 - Pre-mitotic gap phase
 - Synthesis phase
 - M phase
110. During the separation of homologous chromosomes after recombination, X- shaped structures are formed which are called
- Chiasmata
 - Tetrad
 - Recombination nodule
 - Synaptonemal complex
111. Duplication of mitochondria and chloroplasts takes place in _____ of cell cycle.
- G₁ phase
 - G₀ phase
 - S phase
 - G₂ phase
112. Bivalent chromosome can be clearly seen as tetrad in
- Zygotene
 - Leptotene
 - Pachytene
 - Diplotene

113. The best stage to study morphology of chromosomes is

- (1) Prophase
- (2) Anaphase
- ~~(3) Metaphase~~
- (4) Telophase

114.



The event which is associated with the stage represented in the above diagram is

- (1) Condensation of chromatin into chromosome
- (2) Formation of ER and Golgi apparatus
- (3) Alignment of chromosome at the equator
- (4) Splitting of centromere and separation of chromatids.

115. The immediate next lower category to kingdom in taxonomic hierarchy for plants is

- (1) Genus
- ~~(2) Division~~
- (3) Class
- (4) Order

116. Choose the incorrect statement from the following

- (1) Unicellular as well as multicellular organisms grow by cell division
- ~~(2) Reproduction is essential for the survival of an individual~~
- (3) Metabolism is not exhibited in non-living objects
- (4) Living organisms are aware of their surroundings

117. Which of the following is not true about binomial nomenclature?

- (1) Biological names are generally taken from Latin language
- (2) Biological names are printed in italics
- ~~(3) First word denotes the specific epithet~~
- (4) When handwritten, biological name is separately underlined

118. Systematics

- (1) Is derived from a Greek word
- ~~(2) Does not include classification of organisms~~
- (3) Considers taxonomy along with phylogeny
- (4) Does not deal with diversity of organisms

119. Local names of various plants and animals

- (1) Help in recognising organisms worldwide
- (2) Are used universally
- (3) Are specific and distinct names
- (4) Vary from place to place

120. In which of the following options, the taxa do not belong to the same taxonomic category?

- (1) Hominidae and Anacardiaceae
- (2) Diptera and Insecta
- (3) Arthropoda and Chordata
- (4) Primata and Poales

121. Read the following statements and select the correct option.

Statement (A): Lower the taxa, more are the characteristics that the members within the taxon share.

Statement (B): Higher the category, greater is the difficulty of determining the relationship to other taxa at the same level.

- (1) Only statement (A) is correct
- (2) Only statement (B) is correct
- (3) Both statements (A) and (B) are correct
- (4) Both statements (A) and (B) are incorrect

122. All are specific epithets of *Panthera*, except

- (1) leo
- (2) tigris
- ~~(3) sapiens~~
- (4) pardus

123. Which one is odd w.r.t. the processes that are basic to taxonomy?

- (1) Characterisation
- ~~(2) Modification~~
- (3) Classification
- (4) Nomenclature

124. Modern taxonomic studies are based on all, except

- (1) External and internal features
- ~~(2) Ecological informations~~
- (3) Developmental processes
- (4) Phylogeny

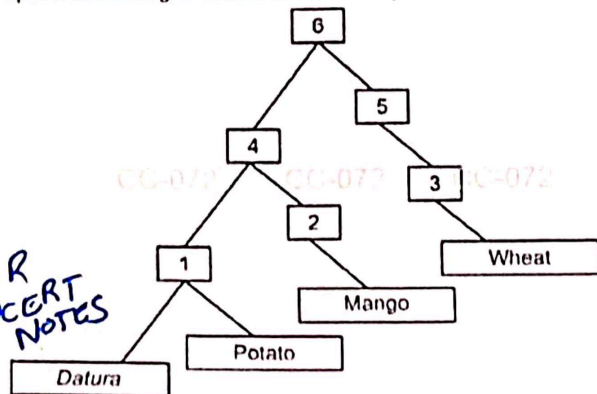
125. Choose the odd one w.r.t. taxonomic categories of housefly.

- ~~(1) Chordata~~
- (2) Muscidae
- (3) Diptera
- (4) Insecta

126. Rules of Binomial nomenclature were given by

- (1) Robert Hooke
- (2) Rudolf Virchow
- ~~(3) Carolus Linnaeus~~
- (4) Ernst Mayr

127. Recognise the following flow diagram and find the correct option according to taxonomic hierarchy



- ~~(1) 1-Polymoniales, 2-Sapindales, 3-Poales, 4-Dicotyledonae, 5-Monocotyledonae, 6-Angiospermae~~
- (2) 1-Solanaceae, 2-Anacardiaceae, 3-Poaceae, 4-Polymoniales, 5-Poales, 6-Angiospermae
- (3) 1-Solanum, 2-Mangifera, 3-Triticum, 4-Angiospermae, 5-monocotyledonae, 6-plantae
- (4) 1-Polymoniales, 2-Sapindales, 3-Poales, 4-Angiospermae, 5-Monocotyledonae, 6-Plantae

128. One of the kingdoms proposed by R.H. Whittaker is divided into two domains in the six kingdom classification. Body organisation of the members of this kingdom is

- (1) Compact tissue type
- (2) Loose tissue type
- (3) Organ system type
- ~~(4) Cellular type~~

129. Archaeobacteria are dissimilar to eubacteria as the former have

- a. Peptidoglycan in their cell wall.
- b. Branched chain lipids in the membrane.
- c. Different sequences of 16S ribosomal RNA genes.

- (1) a and c only
- ~~(2) b and c only~~
- (3) a and b only
- (4) All a, b and c

130. Main criteria for classification used by R.H. Whittaker includes

- (a) Body organisation
- (b) Mode of nutrition
- (c) Phylogenetic relationship
- (d) Simple morphological characters

- (1) (a), (b) and (d) only
- (2) (a) and (b) only
- ~~(3) (a), (b) and (c) only~~
- (4) All (a), (b), (c) and (d)

131. Consider the following statements regarding methanogens and select the correct option.

- A. They are obligate anaerobes
- B. Are responsible for production of a greenhouse gas from dung of ruminant animals
- C. Live in harsh habitats like extreme salty areas
- D. Are chemoautotrophs

- ~~(1) A, B and D are correct~~
- (2) B and D are incorrect
- (3) Only A and B are correct
- (4) B, C and D are correct

132. Match the column I with column II and select the correct option.

Column I	Column II
2a. Methanogens	(i) Photoautotrophic
3b. Halophiles	(ii) Guts of ruminants
4c. Thermoacidophiles	(iii) Heterotrophic
1d. Cyanobacteria	(iv) Hot water springs

- (1) a(ii), b(iii), c(i), d(iv)
- ~~(2) a(ii), b(iii), c(iv), d(i)~~
- (3) a(iii), b(ii), c(i), d(iv)
- (4) a(iii), b(ii), c(iv), d(i)

133. Read the given statements of Assertion (A) and Reason (R) and choose the correct option.

Assertion (A): Bacteria are the sole members of the Kingdom Monera.

Reason (R): Bacteria are the most abundant micro-organisms.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- ~~(2) Both (A) and (R) are true but, (R) is not the correct explanation of (A)~~
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are false

134. The most common method of reproduction in bacteria which occur at favourable temperature, moisture and availability of nutrients is

- (1) Binary fission
- ~~(2) Endospore formation~~
- (3) Conjugation
- (4) Transduction

135. Comma-shaped bacteria are

- (1) Bacillus
- (2) Coccus
- ~~(3) Vibrium~~
- (4) Spirillum

ZOOLOGY

136. Nicotinamide adenine dinucleotide (NAD) and NADP contain the vitamin niacin. Here,

- (1) Niacin acts as the prosthetic group
- (2) NAD acts as a co-enzyme
- (3) NADP is tightly bound to the apoenzyme
- (4) Niacin forms strong coordination bonds with side chains at the active site

137. The structure of protein which provides positional information of amino acids is 'X' that possess

- (1) Hydrogen bonds
- (2) Peptide bonds
- (3) Disulphide bonds
- (4) Ester bonds

138. Read the following statements and choose the correct answer.

Statement A : All prosthetic groups are inorganic compounds that are tightly bound to the apoenzyme.

Statement B : Co-enzymes are organic compounds and their association with the apoenzyme is for longer period of time.

- (1) Both statements (A) and (B) are correct
- (2) Both statements (A) and (B) are incorrect
- (3) Only statement (A) is correct
- (4) Only statement (B) is correct

139. Which of the following does not possess a heterocyclic ring in its structure?

- (1) Adenylic acid
- (2) Cholesterol
- (3) Ribose
- (4) Uracil

140. Consider the following features:

- (a) Enables glucose transport into cells
 - (b) Heteropolymeric structure
 - (c) Belongs to the category of biomacromolecule that is similar to antibodies
- Select the option for which all the above given features hold true.

- (1) Trypsin
- (2) GLUT-4
- (3) Collagen
- (4) Carbonic anhydrase

114-4

149. Select the **incorrect** statement w.r.t. structure of a protein.

- (1) Primary structure gives the positional information of amino acids in a protein
- (2) In a protein, only left-handed helices are observed.
- (3) Tertiary structure gives a 3-dimensional view of a protein
- (4) Haemoglobin shows optimal activity at quaternary level of organisation

150. Cholesterol is essential component of cell membrane of animals. Cholesterol is an example of a

- (1) Simple lipid
- (2) Compound lipid
- (3) Derived lipid
- (4) Conjugated lipid

151. The EC number of an enzyme is 3.7.1.1. To which main class of enzymes does it belong?

- (1) Hydrolases
- (2) Lyases
- (3) Ligases
- (4) Transferases

152. In primary structure of a protein, methionine is located at left end of polypeptide chain while glutamic acid is located at right end. Which of the following statements is true with respect to above mentioned protein?

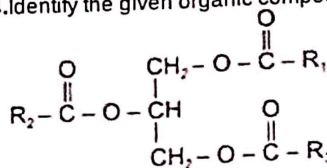
- (1) Methionine is a C-terminal amino acid
- (2) Glutamic acid is the first amino acid of the chain
- (3) Glutamic acid is N-terminal amino acid
- (4) Methionine is the first amino acid of the chain

153. Fats have A melting point and remain as B at room temperature while oils have C melting point and remain as D at room temperature.

	A	B	C	D
(1)	High	Liquid	Low	Solid
(2)	Low	Liquid	High	Solid
(3)	Low	Solid	High	Liquid
(4)	High	Solid	Low	Liquid

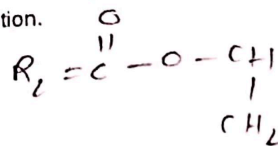
- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

154. Identify the given organic compound.



Choose the **correct** option.

- (1) Triglyceride
- (2) Diglyceride
- (3) Monoglyceride
- (4) Cholesterol



155. How many fatty acid residue(s) are present in a lecithin molecule?

- (1) 1
- (2) 2
- (3) 3
- (4) 4

156. A competitive inhibitor of enzyme succinic dehydrogenase is

- (1) Malonate
- (2) Fumarate
- (3) Succinate
- (4) Oxalate

157. Assertion(A): In competitive inhibition of enzymes, the inhibitor structurally resembles to the substrate.
Reason (R): Each enzyme is not specific for its substrate in a chemical reaction.

In the light of above statements, select the correct option.

- (1) Both (A) and (R) are false
- (2) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

158. Uridine differs from uridylic acid in

- (1) Absence of phosphate moiety
- (2) Presence of uracil
- (3) Presence of pentose sugar
- (4) Presence of phosphate

159. Which one is an **incorrect** description of apoenzyme?

- (1) Organic compound
- (2) Proteinaceous in nature
- (3) Part of holoenzyme
- (4) Complete enzyme

170. Which of the following is not included in conducting part of respiratory system?
- Trachea
 - Nasal cavity
 - Pharynx
 - ~~Alveoli~~
171. During normal inspiration, pressure gradient between the atmosphere and the alveoli is created with the help of specialised muscles. Select the correct set of such muscles.
- External inter-costal muscles
 - Diaphragm
 - Abdominal muscles
- a, b and c
 - a and b
 - b and c
 - a and c
172. Inspiration in humans is brought about by
- Contraction of diaphragm and relaxation of external intercostal muscles
 - Relaxation of diaphragm and contraction of external intercostal muscles
 - Relaxation of diaphragm and relaxation of internal intercostal muscles
 - Contraction of both diaphragm and external intercostal muscles
173. The air that we breathe in and the food that we ingest, travel through various tubes and organs as they move along the body. A portion of which of the following structures is involved in the conduction of both food as well as air?
- Pharynx
 - Oesophagus
 - Larynx
 - Trachea
174. The average volume of air inhaled by an individual in a normal inhalation is
- ~~500 mL~~
 - 5 mL
 - 15 mL
 - 15 L
175. The first step of respiration is
- Transport of gases from lungs to blood
 - Transport of gases from blood to tissues
 - ~~Exchange of O_2 from the atmosphere with CO_2 rich alveolar air~~
 - Catabolism of food substances to release energy within cells
176. Thoracic chamber is ventrally, dorsally and laterally formed respectively by
- Vertebral column, ribs, and sternum
 - ~~Ribs, sternum, and diaphragm~~
 - Sternum, vertebral column, and ribs
 - Diaphragm, sternum, and ribs
177. Ciliated epithelium lining the bronchioles helps in
- Exchange of gases
 - Reabsorption of water
 - Filtration of blood
 - Movement of mucus and trapped particles in a specific direction through respiratory tract
178. A catabolic biological process in living organisms to break organic molecules to release energy is known as
- Digestion
 - ~~Respiration~~
 - Excretion
 - Combustion
179. Which is not a function of conducting part of respiratory tract?
- Trapping dust particles
 - Humidification of incoming air
 - Adjusting temperature of air
 - ~~Trapping O_2 in blood~~
180. In humans, route of passage of air from lungs to atmosphere is
- ~~Alveoli → Bronchioles → Bronchi → Trachea → Larynx → Pharynx → Nasal cavity~~
 - Alveoli → Bronchi → Bronchioles → Trachea → Pharynx → Larynx → Nasal cavity
 - Alveoli → Bronchioles → Bronchi → Trachea → Pharynx → Larynx → Nasal cavity
 - Alveoli → Bronchi → Bronchioles → Trachea → Larynx → Pharynx → Nasal cavity

160. Match column I and column II w.r.t. respiratory volumes in mL.

Column I		Column II
a. Tidal volume	(i)	1100-1200
b. Inspiratory reserve volume	(ii)	1000-1100
c. Expiratory reserve volume	(iii)	2500-3000
d. Residual volume	(iv)	500

Choose the correct option.

- (1) a(i), b(ii), c(iii), d(iv)
- (2) a(ii), b(iii), c(iv), d(i)
- (3) a(iii), b(iv), c(i), d(ii)
- (4) a(iv), b(iii), c(ii), d(i)

161. Assertion (A) : Mechanism of breathing varies among different groups of animals depending on their habitats and levels of organisation.

Reason (R) : Respiration involves diffusion of O_2 and CO_2 between blood and tissue based on concentration gradient. In the light of above statements, select the most appropriate option.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are false

162. In humans, all of the following events occur during expiration, except

- (1) Occurrence of negative pressure in the lungs w.r.t. atmospheric pressure
- (2) Volume of thoracic cavity decreases
- (3) Relaxation of external inter-costal muscles
- (4) Relaxation of diaphragm causing it to become dome-shaped.

163. Trachea divides at the level of A to form right and left primary B.

Choose the option that fills A and B correctly.

- (1) 5th thoracic vertebra, bronchi
- (2) 3rd thoracic vertebra, bronchi
- (3) 5th thoracic vertebra, bronchioles
- (4) 3rd thoracic vertebra, bronchioles

164. Read the following features:

- i. Presence of thin flattened simple squamous epithelium
- ii. Absence of cilia
- iii. Absence of cartilage

Choose the option which fulfills all the above criteria.

- (1) Primary bronchi
- (2) Pharynx
- (3) Alveoli
- (4) Trachea

165. Choose the incorrect statement w.r.t. human sound box.

- (1) It is a cartilaginous structure
- (2) It is also called larynx
- (3) It is common passage for food and air
- (4) It is covered by a cartilaginous flap, epiglottis, during swallowing

166. The total volume of air present in the lungs after maximum inspiration is

- (1) $ERV + TV + IRV$
- (2) $RV + ERV + TV + IRV$
- (3) $IC - FRC$
- (4) $IC + ERV$

167. Read the following statements and choose the correct option.

Statement A: O_2 is utilized by the organisms to indirect breakdown of simple molecules like glucose, amino acids etc., to derive energy to perform various activities.

Statement B: CO_2 is non-harmful and is released during the catabolic and anabolic reactions.

- (1) Both statements A and B are incorrect
- (2) Both statements A and B are correct
- (3) Only statement A is correct
- (4) Only statement B is correct

168. Complete the analogy.

Aquatic molluscs : Gills :: Earthworm : X

Choose the correct option for X.

- (1) Trachea
- (2) Moist cuticle
- (3) Book lungs
- (4) Book gills

169. The rate of breathing in a healthy human adult under normal physiological conditions is

- (1) 38-44 times/min
- (2) 12-16 times/min
- (3) 70-72 times/min
- (4) 20-24 times/min